STOOGE · STIG TRACKING, OVERSIGHT, AND ORGANIZATIONAL GOVERNANCE ENGINE · EROMAN53 OPENRMF FORK · JULY 2026

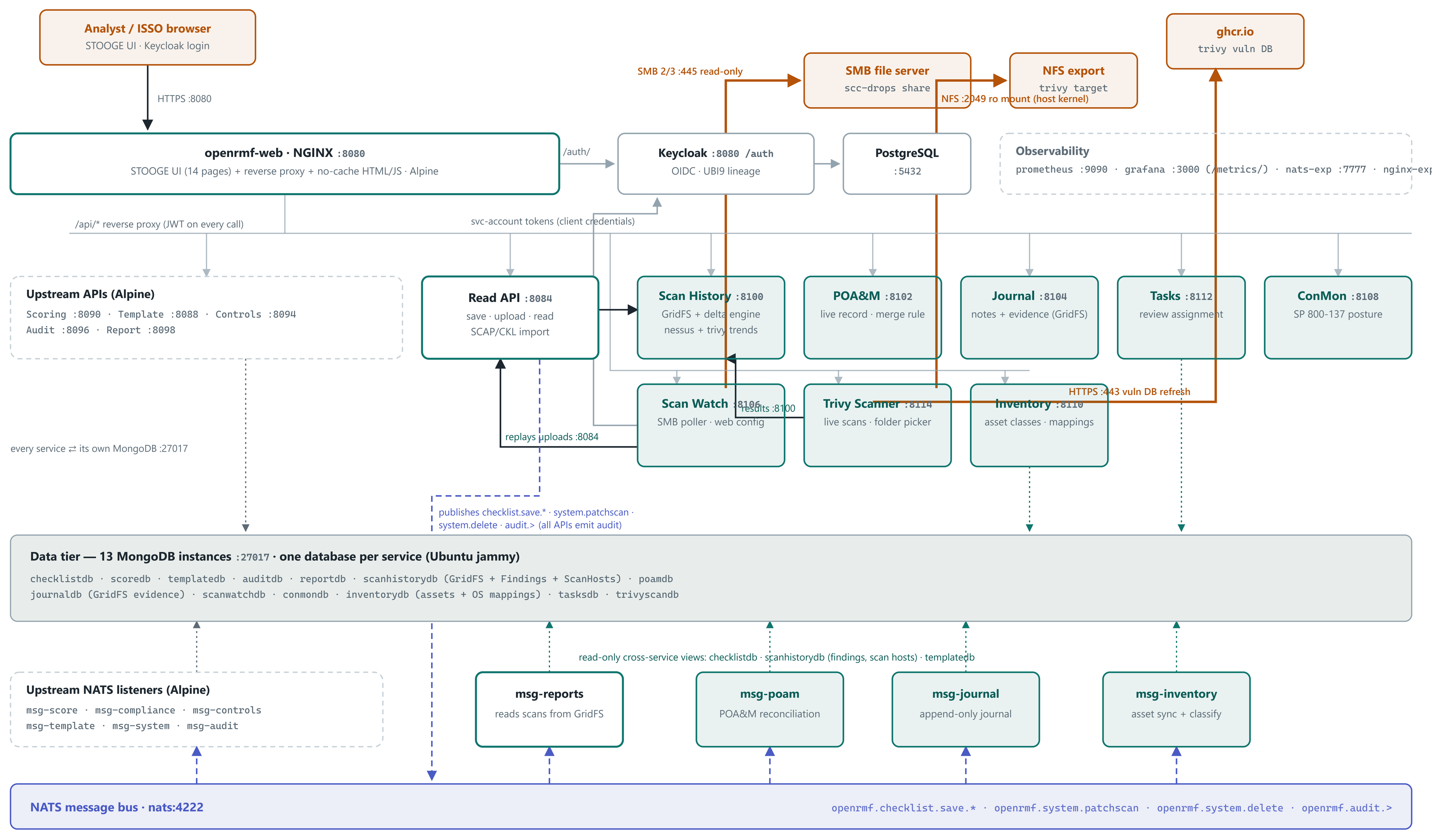
Architecture & Communication Ports / Protocols

Every box is a container on the internal `openrmf` Docker network. NGINX is the single host-exposed entry point; the NATS bus is the service backbone; every service owns exactly one MongoDB. All fork-built .NET services now run on the **UBI8 (RHEL 8.10) base** with the DoD CA bundle. Teal marks fork additions/modifications.

46 containers · **15** HTTP services · **10** NATS listeners · **13** Mongo DBs · **1** host-exposed port (8080) · UBI8 fork services

1 · Container architecture

upstream (Alpine) · fork-modified (UBI8) · fork-new (UBI8) · external | HTTP / REST · NATS pub/sub · MongoDB · SMB / NFS / registry (boundary)



All fork services (teal) run on eroman53/openrmf-base-ubi:8.10 — ubi8-minimal + DoD CA bundle, curl/glib2 stripped, 0 CRIT / 1 HIGH per image.
Test-only samba container (local override) excluded. Keycloak login theme still upstream-branded.

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2 · Communication ports & protocols (PPSM view)

INBOUND (HOST BOUNDARY)

1 port

`8080/tcp` HTTP — the only host-published port; UI, all APIs, and OIDC ride through NGINX. TLS on `8443` is provisioned (commented) in the config.

OUTBOUND (HOST BOUNDARY)

3 flows

`445/tcp` SMB (scan drops, read-only) · `2049/tcp` NFS (trivy target, read-only; `+111` for v3) · `443/tcp` HTTPS to ghcr.io (Trivy DB; air-gap: pre-seed the cache volume)

INTERNAL ONLY

everything else

All service, database, bus, auth, and metrics traffic stays on the internal `openrmf` Docker network — nothing below the gateway is host-published.

SOURCE	DESTINATION	PORT	PROTOCOL / SERVICE	PURPOSE
BOUNDARY-CROSSING				
Analyst / ISSO browser	openrmf-web (NGINX)	`8080/tcp`	HTTP (TLS 8443 available)	UI pages, all `/api/*` calls, OIDC login redirects
openrmf-svc-scanwatch	SMB file server	`445/tcp`	SMB 2/3	Polls the scan drop share (read-only account)
Docker host kernel	NFS server	`2049/tcp (+111 v3)`	NFS v4 (v3 opt.)	Read-only mount of the Trivy scan target
openrmf-svc-trivy	ghcr.io	`443/tcp`	HTTPS / OCI	Trivy vulnerability DB refresh (cached in volume; skippable air-gapped)
GATEWAY (NGINX REVERSE PROXY)				
NGINX	Keycloak	`8080/tcp`	HTTP / OIDC	`/auth/` — login, tokens, JWKS
NGINX	Grafana	`3000/tcp`	HTTP	`/metrics/` dashboards
NGINX	Read API	`8084/tcp`	HTTP REST	`/api/read/` save · upload · read
NGINX	Template / Scoring / Controls / Audit / Report APIs	`8088 · 8090 · 8094 · 8096 · 8098/tcp`	HTTP REST	Upstream API routes
NGINX	Scan History / POA&M / Journal / Scan Watch / ConMon / Inventory / Tasks / Trivy Scanner	`8100 · 8102 · 8104 · 8106 · 8108 · 8110 · 8112 · 8114/tcp`	HTTP REST	Fork service routes (`/api/scanhistory/ · /api/poam/ · /api/journal/ · /api/scanw`
IDENTITY & AUTH				
Every API / service	Keycloak	`8080/tcp`	HTTP / OIDC	Issuer discovery + JWKS for JWT validation on every request
Scan Watch · Trivy Scanner	Keycloak	`8080/tcp`	HTTP / OIDC token	Client-credentials tokens (service account `openrmf-scanwatch`, Editor+Rea
Keycloak	PostgreSQL	`5432/tcp`	PostgreSQL	Realm / user store
SERVICE-TO-SERVICE HTTP				
Read API	Scan History API	`8100/tcp`	HTTP REST	Streams .nessus uploads into GridFS + delta engine
Scan Watch	Read API	`8084/tcp`	HTTP REST	Replays dropped SCC XCCDF / CKL / Nessus files as authenticated uploads
Trivy Scanner	Scan History API	`8100/tcp`	HTTP REST	Uploads Trivy JSON as `scanType=trivy` snapshots
MESSAGING BACKBONE				
Read API (+ every API for audit events)	NATS	`4222/tcp`	NATS pub	`openrmf.checklist.save.* · system.patchscan · system.delete`
10 msg listeners (score · compliance · controls · template · system · audit · reports · poam · journal · inventory)	NATS	`4222/tcp`	NATS sub	Event-driven reconciliation, journaling, scoring, reporting
DATA TIER				
Each service	Its own MongoDB (13 instances)	`27017/tcp`	MongoDB wire	One database per service — no shared writes
msg-poam · msg-reports · msg-journal · msg-inventory · ConMon · Inventory API	checklistdb · scanhistorydb · templatedb · poamdb	`27017/tcp`	MongoDB wire	Deliberate read-only cross-service views (findings, scan hosts, checklists, tem
OBSERVABILITY				
Prometheus	Service `/metrics` endpoints · NATS exporter · NGINX exporter	`service ports · 7777 · 9113/tcp`	HTTP scrape	Metrics collection across every tier
NATS exporter	NATS monitoring	`8222/tcp`	HTTP	varz / connz / subz
Grafana	Prometheus	`9090/tcp`	HTTP / PromQL	Dashboards (served to users via NGINX `/metrics/`)

Sources: `openrmf-docs/scripts/docker-compose.yml` + `nginx.conf` as of the UBI8 rebase (2026-07-18). Ports verified against the live nginx upstream config; all fork images scan 0 CRITICAL / 1 accepted HIGH. This matrix doubles as the starting point for the Phase 2 automated PPSM module.